Examiner only

(c) Tachyhydrite is a soft, soluble mineral that has a general formula of $A_aB_bX_c.dH_2O$. A and B represent two different s-block metal ions, with X representing an anion.

Use **all** the following data to find the identity of all the ions present and to find the formula of tachyhydrite.

A, B and X are not the chemical symbols of the elements present.

A flame test on a sample of tachyhydrite gives a brick-red flame with no indication of any other colour. Addition of dilute sulfuric acid to a solution of tachyhydrite causes the solution to go cloudy. With concentrated sulfuric acid, misty fumes are released from the solid, however there is no colour change. Addition of aqueous silver nitrate to a solution of tachyhydrite causes a precipitate to form.

Tachyhydrite has an M_r of approximately 518. On heating to constant mass the solid loses 41.75 % of its initial mass. [6 QER]

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Formula of tachyhydrite

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